



Executive Value Proposition: In software engineering, code diffs and PR reviews are instantaneous. In hardware engineering, teams still rely on manual screenshots, Word documents, and red pen markups. A single unnoticed PCB routing or pin assignment error causes a factory fabrication re-spin costing **\$2,000 to \$20,000+ and 3 to 4 weeks in project delay**. Banana 2.0 eliminates this risk with automated visual Git diffs, interactive overlay sliders, and semantic audit tracking.

1. COMMERCIAL PRICING TIERS

COMMUNITY

Free

\$0

For hobbyists, students, and open-source hardware creators.

- ✓ Unlimited public GitHub repos
- ✓ Drag & drop comparisons (3/mo)
- ✓ Side-by-side visual diff
- ✓ Overlay slider & delta maps
- ✓ Pan & zoom vector navigation
- ✓ KiCad 7, 8, 9 & 10 support

SOLO ENGINEER

Pro

\$19 / mo

For freelance electrical engineers and hardware consultants.

- ✓ **Unlimited private repos**
- ✓ Unlimited drag-and-drop diffs
- ✓ **Full Semantic Audit Log**
- ✓ Net displacement & trace tracking
- ✓ Export Design Review PDFs
- ✓ Client shareable view links
- ✓ Standard email support

STARTUPS & LABS

Team

\$49 / seat / mo

For robotics, IoT, and electronics engineering teams (min 3 seats).

- ✓ **GitHub & GitLab PR Bot**
- ✓ **Pin-point review comments**
- ✓ PR approval status gates
- ✓ Cross-layer difference filters
- ✓ Team roles & permissions
- ✓ Priority serverless render queue
- ✓ Slack / Discord notifications

DEFENSE & MEDTECH

Enterprise

\$2,500+ / yr

For aerospace, automotive, medical, and defense contractors.

- ✓ **On-Prem / VPC Docker Runner**
- ✓ **Zero IP leaves client network**
- ✓ ITAR & NDA compliance ready
- ✓ SAML SSO (Okta, Azure AD)
- ✓ Custom audit logs & retention
- ✓ Dedicated account manager
- ✓ 99.9% Uptime SLA

2. DETAILED FEATURE MATRIX

Feature / Capability	Free	Pro (\$19)	Team (\$49/seat)	Enterprise
Public GitHub Repositories	Unlimited	Unlimited	Unlimited	Unlimited
Private Repositories & Local Uploads	3 / month	Unlimited	Unlimited	Unlimited
Visual Modes (Side-by-Side, Slider, Delta)	✓	✓	✓	✓
High-Speed Vector Diffing (Sub-second)	✓	✓	✓	✓
Semantic Audit Log & Net Identification	—	✓	✓	✓
Automated GitHub / GitLab PR Bot	—	—	✓	✓
Collaborative PCB Comment Pins	—	—	✓	✓
Design Review Sign-off PDF Export	—	✓	✓	✓
PR Merge Gates & Status Checks	—	—	✓	✓
Self-Hosted VPC Runner (Zero Cloud Storage)	—	—	—	✓
SAML SSO & Custom Security Controls	—	—	—	✓

The "Killer Feature": PR Visual Bot

When an engineer opens a GitHub Pull Request modifying .kicad_pcb, the Banana Bot comments with a rich interactive summary:

- **Banana Bot:** Detected 4 modifications:
 - Changed SPI2_CS trace (+0.4mm spacing)
 - Added C14 bypass capacitor on 3V3 rail

[\[Click to Review Interactive Visual Diff →\]](#)

Intellectual Property & Security

Hardware schematics are proprietary trade secrets. Banana addresses this through a tiered security architecture:

- **Ephemeral Cloud Workers:** Raw files processed in isolated memory and discarded immediately after rendering.
- **On-Premise VPC Runners:** For defense & healthcare, Docker runners execute inside the client's AWS/GCP perimeter.



3. CLOUD DEPLOYMENT ARCHITECTURE

Banana 2.0 achieves **93%+ gross margins** by decoupling static edge frontend delivery from serverless containerized compute that automatically scales to zero when idle.

🌐 Edge Frontend Layer (Cloudflare Pages)

- **Global CDN Hosting:** React + Vite frontend distributed across 300+ global edge nodes.
- **Instant Zero-Latency Loading:** Sub-50ms initial load times globally.
- **Bandwidth Cost: \$0 / month** (Included in Cloudflare's unlimited bandwidth tier).
- **Custom Domains & SSL:** Automated SSL for app.banana-diff.com.

⚡ Serverless Compute Layer (Google Cloud Run / Fly.io)

- **Containerized KiCad CLI:** Ubuntu Linux Docker image with native kicad package pre-installed.
- **Scale-to-Zero:** Zero CPU or RAM costs during idle periods.
- **High-Speed Render:** 2 vCPU / 2GB RAM container executes KiCad SVG exports in **0.8 to 1.5 seconds**.
- **Deterministic Caching:** SHA-256 disk cache prevents redundant re-rendering of unchanged commits.

4. UNIT ECONOMICS & MARGIN ANALYSIS

Metric	Cost / Value	Explanation
Cloudflare Pages Hosting	\$0.00 / mo	Free edge CDN distribution with unlimited requests.
Compute Cost per Diff Execution	~\$0.00012	2 vCPU running for 1.2s on Google Cloud Run (\$0.00002400/vCPU-sec).
Average Monthly Compute per Active User	~\$0.048 / mo	Assuming an active engineer triggers 400 diff comparisons per month.
Storage & Database (Supabase / Cloudflare R2)	~\$5.00 / mo	Diff vector cache and team authorization records.
Gross Margin on Pro Plan (\$19/mo)	97.4%	Revenue \$19.00 — Infrastructure cost ~\$0.50.
Gross Margin on Team Plan (\$49/seat/mo)	96.8%	Revenue \$49.00 — Infrastructure & API cost ~\$1.55.

5. 12-MONTH FINANCIAL PROJECTIONS (ARR TRAJECTORY)

Growth model based on viral GitHub bot adoption across hardware hackathons, robotics incubators, and KiCad community channels:

Milestone	Pro Users	Teams (Avg 4 seats)	MRR	ARR
Month 3 (Beta Launch)	40	5 (\$980)	\$1,740	\$20,880
Month 6 (GitHub App Launch)	150	25 (\$4,900)	\$7,750	\$93,000
Month 9 (Enterprise Expansion)	350	70 (\$13,720)	\$20,370	\$244,440
Month 12 (Scale)	750	160 (\$31,360)	\$45,610	\$547,320

📈 Viral Growth Mechanics

- **Free PR Billboard:** Every automated GitHub PR comment introduces Banana to reviewers, firmware devs, and managers.
- **Open Source Goodwill:** Free tier for open-source hardware builds deep developer loyalty (like Altium's opposite reputation).
- **Fabrication Partnerships:** Co-marketing opportunities with fab houses (JLCPCB, PCBWay, MacroFab) offering "Verify with Banana" before ordering.
- **Zero Software Lock-in:** Works directly with native KiCad files without proprietary project formats.

6. IMPLEMENTATION ROADMAP (30-DAY MVP PLAN)

PHASE 1: WEEK 1-2

Containerization & Deploy

- Package backend in Docker with Ubuntu KiCad CLI.
- Deploy to Google Cloud Run.
- Connect Cloudflare Pages to frontend.

PHASE 2: WEEK 3

GitHub OAuth & Webhooks

- Add "Login with GitHub".
- Repository picker & branch selector.
- Webhook listener for automated PR diff comments.

PHASE 3: WEEK 4

Billing & Public Launch

- Integrate Stripe Customer Portal & checkout.
- Launch on Product Hunt, Hacker News, & r/KiCad.
- Onboard first 50 beta teams.